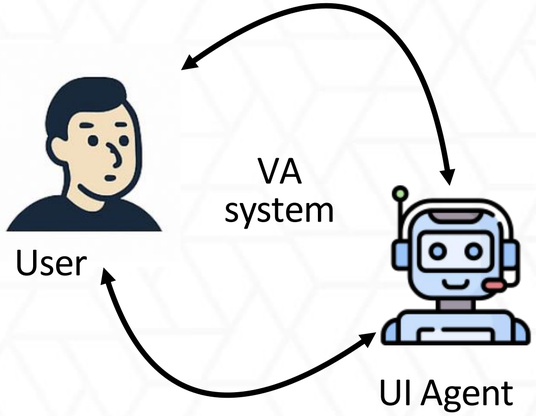
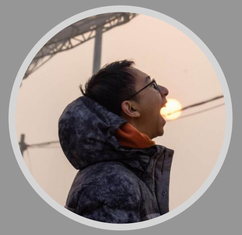
FDUVIS Lab

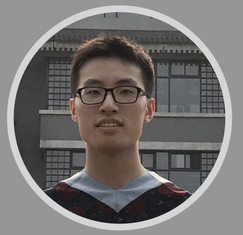
VIS2025



ProactiveVA: Proactive Visual Analytics with LLM-based UI Agent





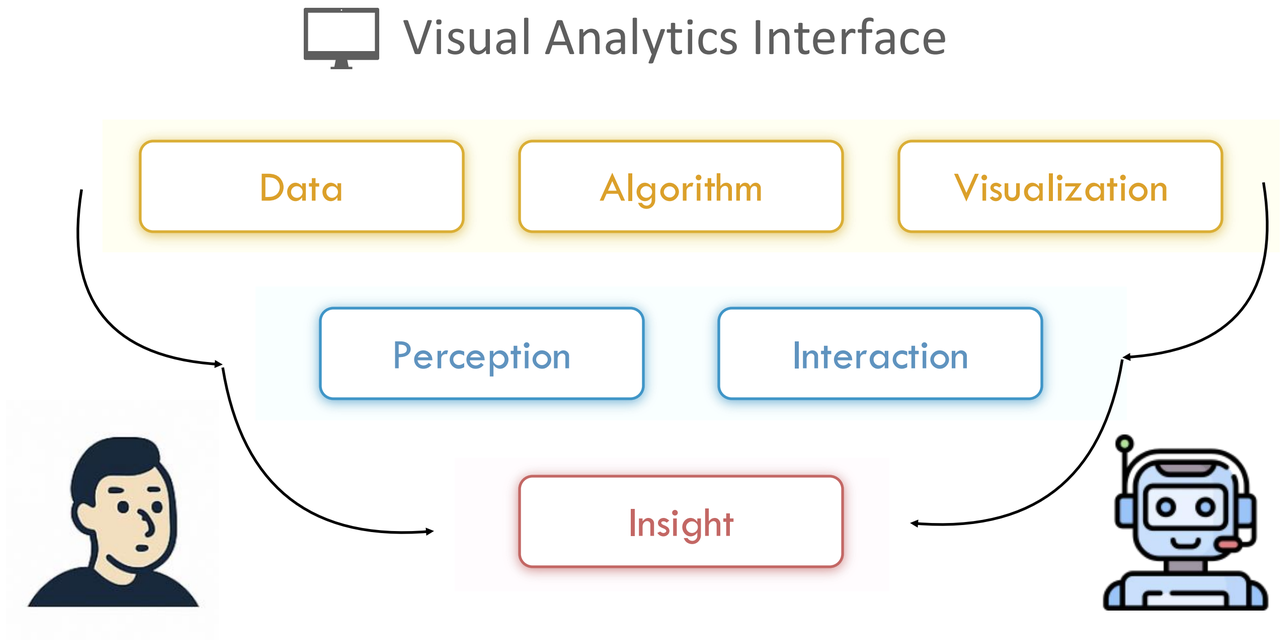



Yuheng Zhao[1], Xueli Shu[1], Liwen Fan[1], Lin Gao[1], Yu Zhang[2], Siming Chen[1]

[1] Fudan University [2] University of Oxford

Background

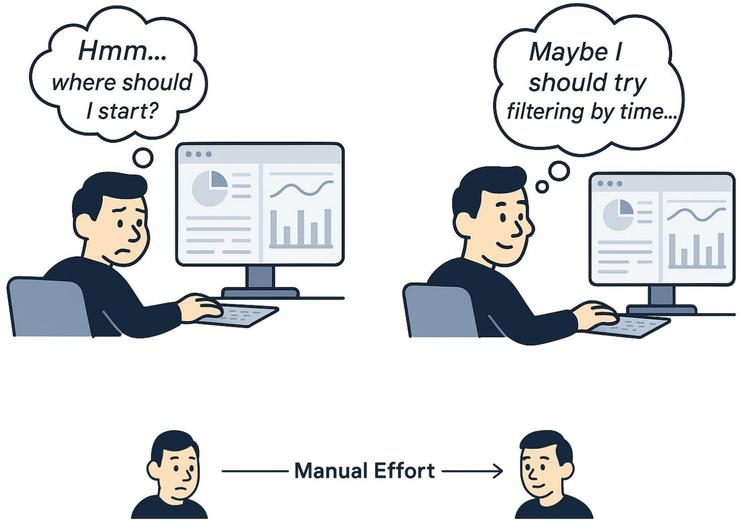
The vision and goal of visual analytics is:



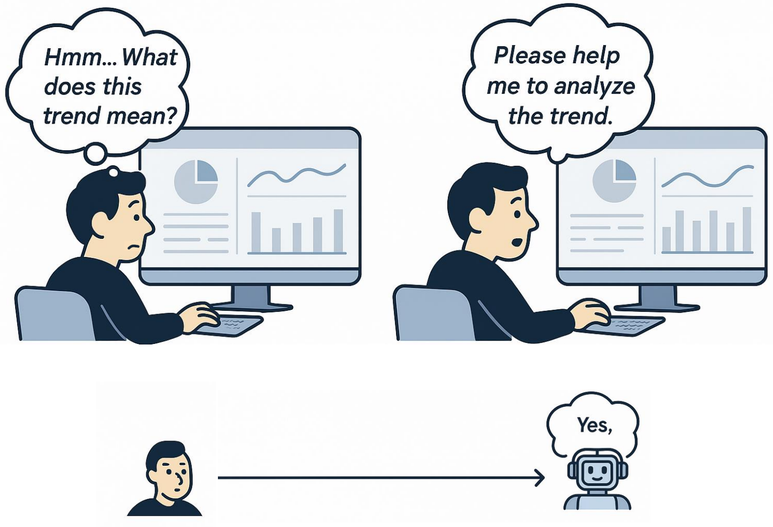
To enable intelligent Human-AI Collaboration in analyzing and understanding data

Comparison of Three Paradigms of Visual Analytics

Traditional Visual Analytics



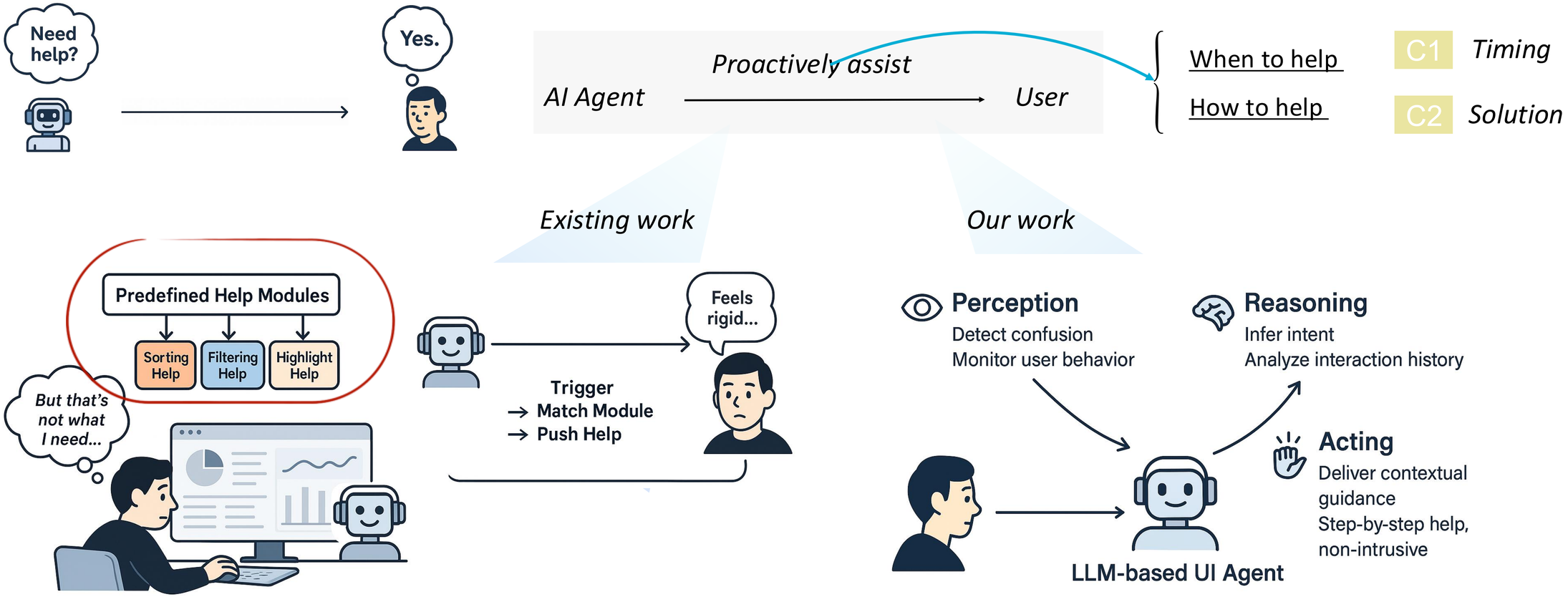
Reactive Visual Analytics



Proactive Visual Analytics



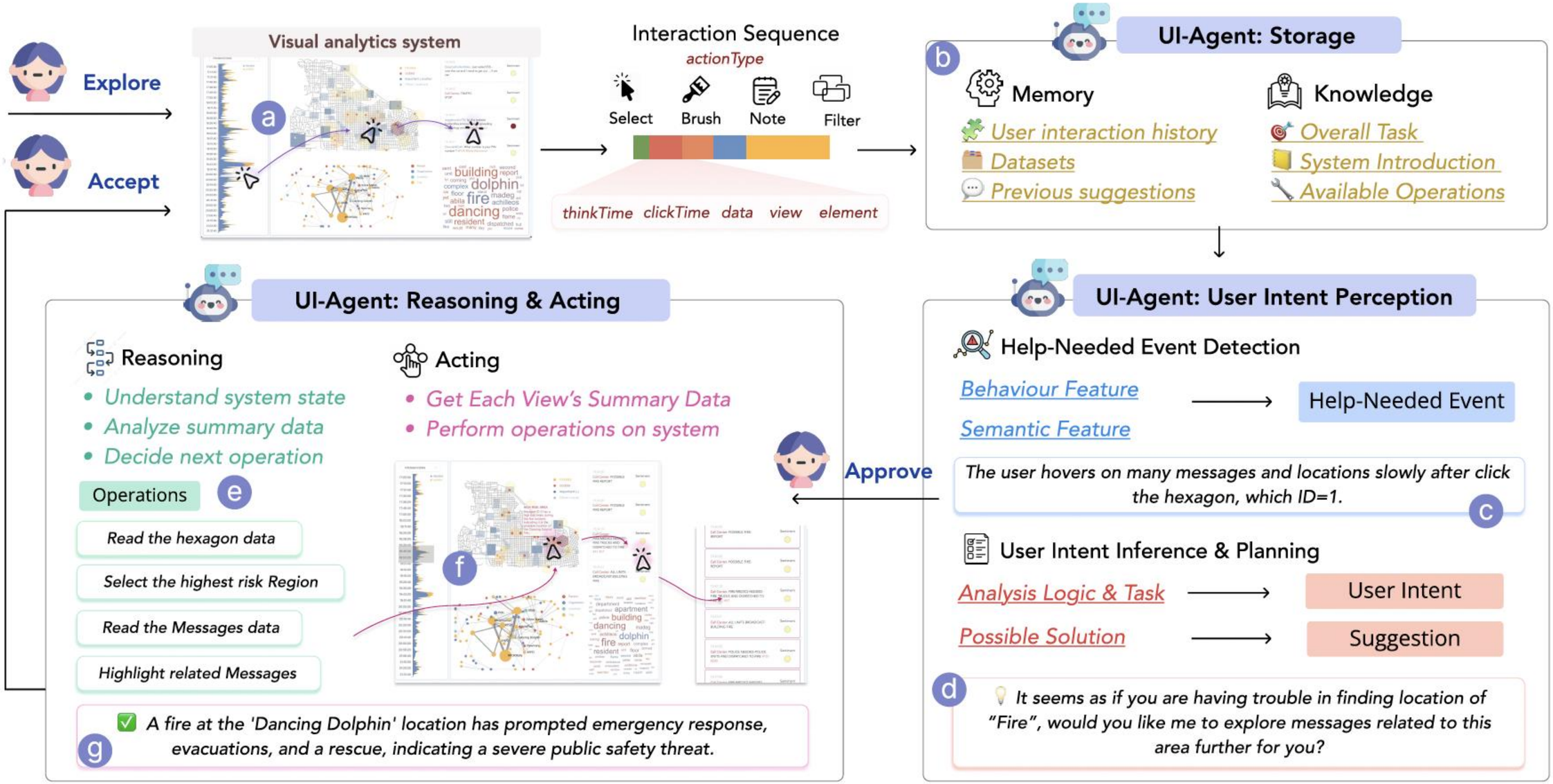
The challenges of proactive agent assisted VA



The timing and solution are rely on pre-defined rules

Autonomously decides when and how to assist

Overall Pipeline of ProactiveVA: “perception - reasoning - acting”



Formative study

Interview –> Log analysis

- Collected exploration data from 10 users (think-aloud, video, interaction logs)
- Developed an interactive visualization tool to annotate the interaction segments
- Summarized three typical help-need situations, and judgment methods to guide the model



	Interaction pattern	Interpretation	Problem Subcategory	Problem Category	Assistance Category	Assistance Sub-Category
1		Hover the message list by frequently scrolling	Unfamiliar with data transformation (3)	Unfamiliar with System Functionality (23)	Onboarding Tips	Provide explanation of data transformation
2		Paused for a long time at node in graph view	Unfamiliar with visual mapping (6)			Provide explanation of visual design
3		Select a keyword but not going to timeline	Unfamiliar with interaction method (14)			Provide explanation of data transformation
4		Select the next keyword hesitantly	Confusion about data meaning (34)	Lack of Understanding of Data (53)	Exploration Guidance	Summarize data meaning from context
5		Browse messages without further action	Hard to find detail from large data (11)			Identify important feature from data
6		Compare different time range slowly	Hard to compare data relation (8)			Compare data to find similarities/diffferences
7		Review the timeline repeatedly	Incompleteness in findings (17)	Failure to Complete Task (29)	Result Verification	Remind incompleted steps
8		Explore the previous analysis path repeatedly	Contradiction in findings (3)			Check conflicting of submitted notes
9		Identify details by switching views slowly	Inaccuracy in findings (9)			Verify findings from data.

Implementation

Stage1 User Intent Perception

1. Help-Needed Event Detection*:

The user is having trouble in finding details of “Fire”

➢Temporal Features: prolonged pauses

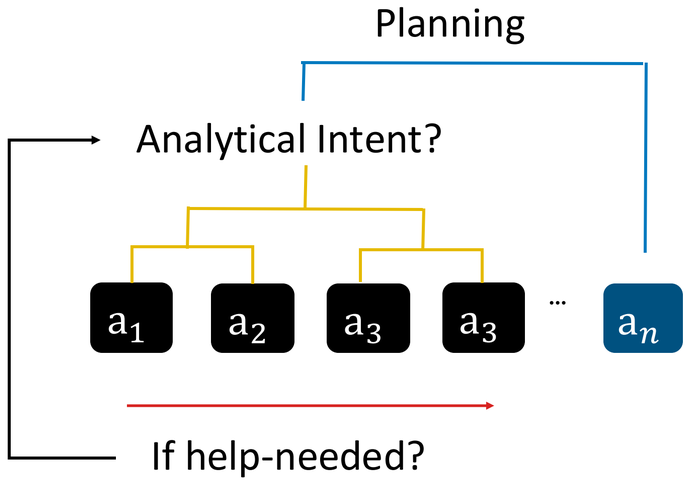
➢Action Features: repetitive or wrong actions

➢Contradictory or Inaccurate notes: compared with previous notes and confirmed by data

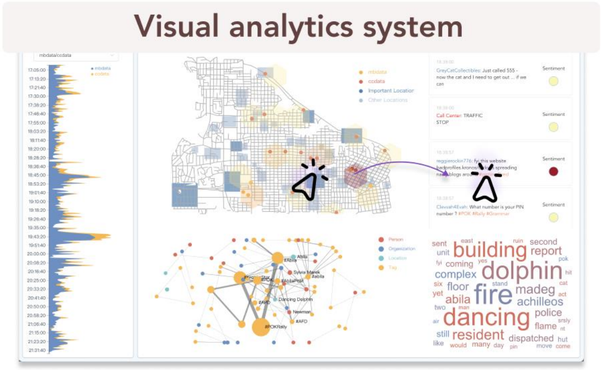
2. Planning

- Infer user’s analytical intent
- propose an initial plan

I need to filtering high-risk regions and reading the related messages



The user hovers on many messages and locations slowly after click the hexagon, which ID=1.



* Gotz, David, and Michelle X. Zhou. "Characterizing users' visual analytic activity for insight provenance." Information Visualization 8.1 (2009): 42-55.

Implementation

Stage2 Reasoning and Acting *

1. Reasoning: Interactive exploration

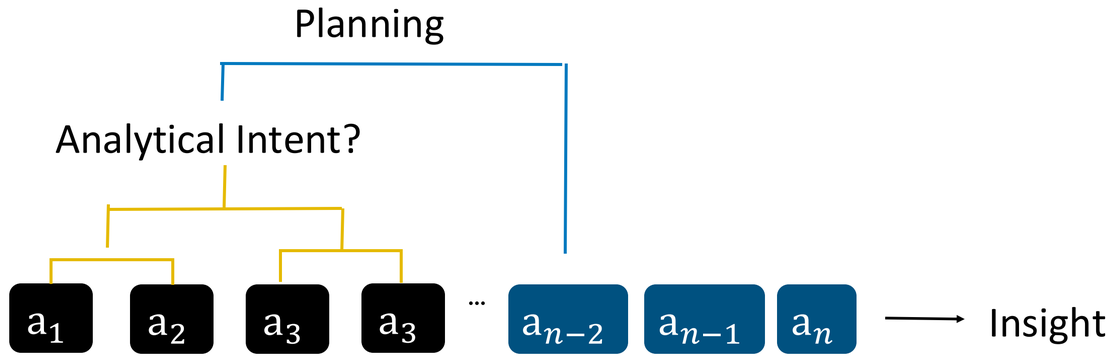
Function	Parameters	Description
`readData`	`view` : Enum (Total Sales, SaleMap, etc.)	Reads data from the specified view.
`select`	`view` : Enum, `field` : Enum, `value` : Array/Object	Selects data based on the view, field, and value.
`filter`	`fieldName` : Enum, `value` : Array/Object	Applies filter across all views.
`finish`	None	Finalizes the operation and completes the task.

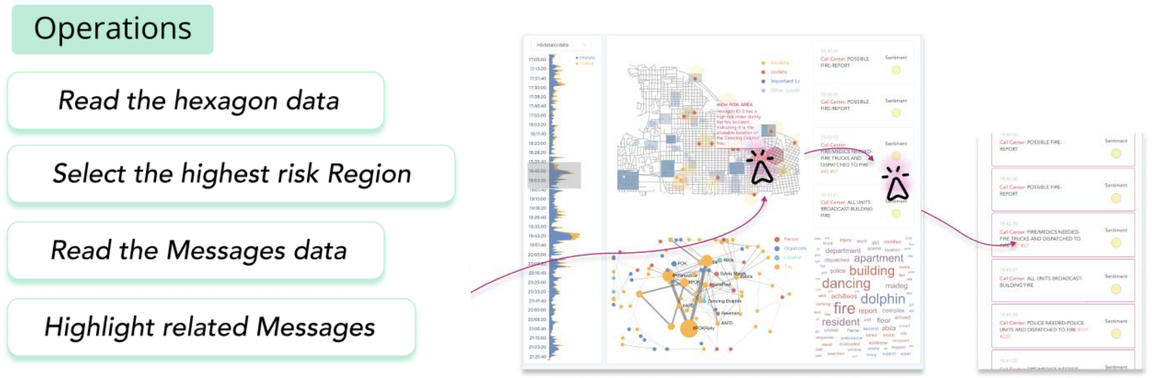
- Input: initial plan, data tables
- Output: observation, next operation, insight

2. Acting

- Components that interact with the system, e.g, getData, applyOperations

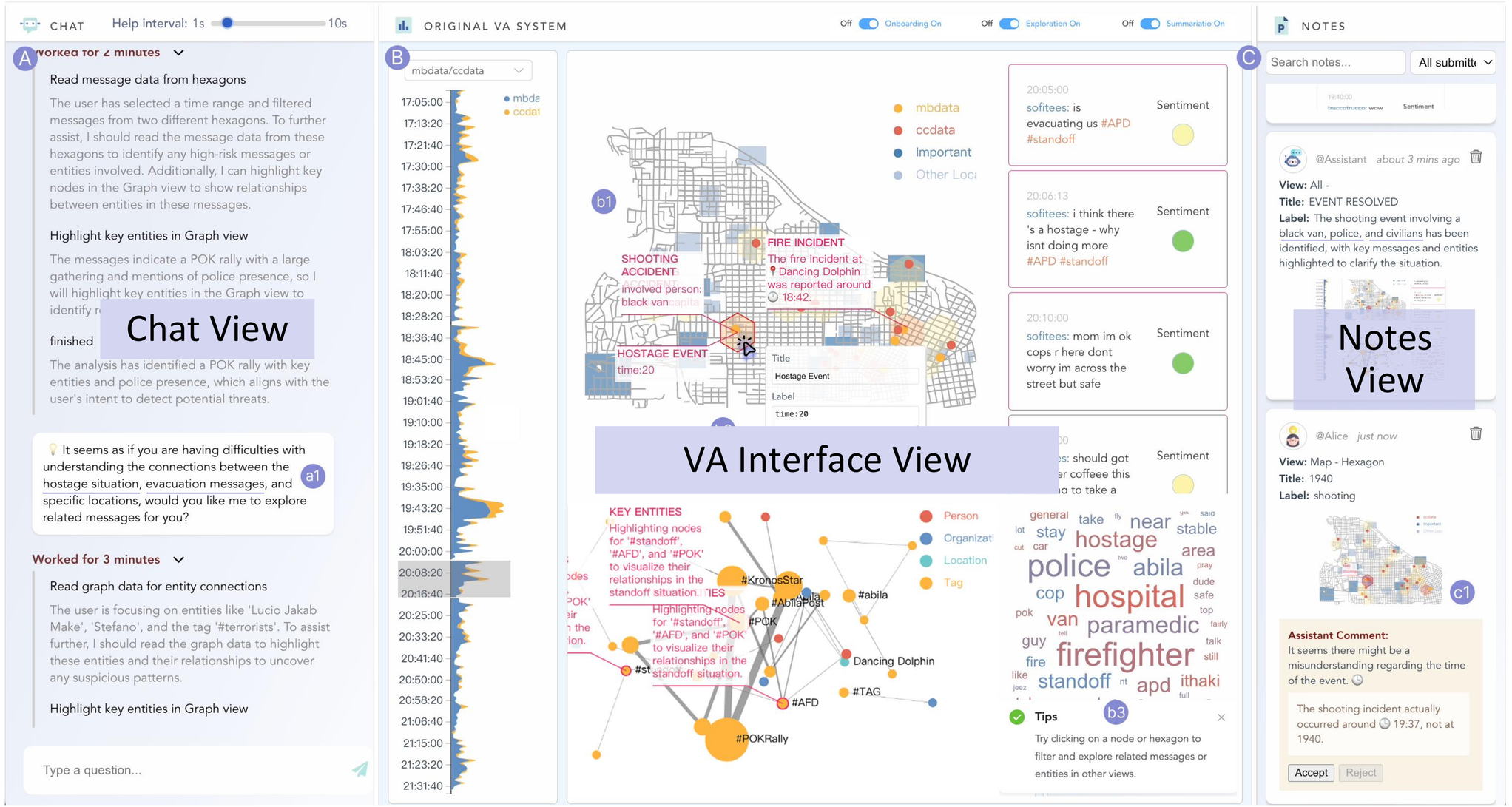
I need to filtering high-risk regions and reading the related messages



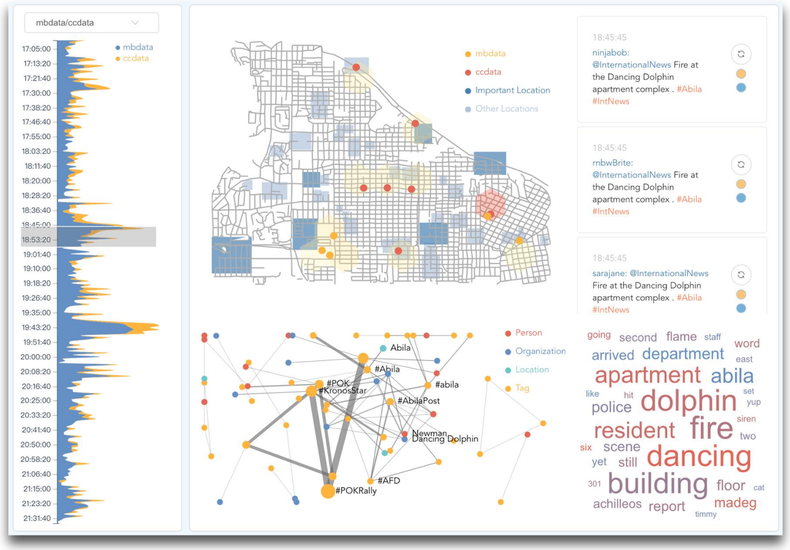


** Yao, Shunyu, et al. "ReAct: Synergizing reasoning and acting in language models." International Conference on Learning Representations (ICLR). 2023.*

Proactive UI Agent-assisted VA Interface



Implement ProactiveVA in Two Types of VA System



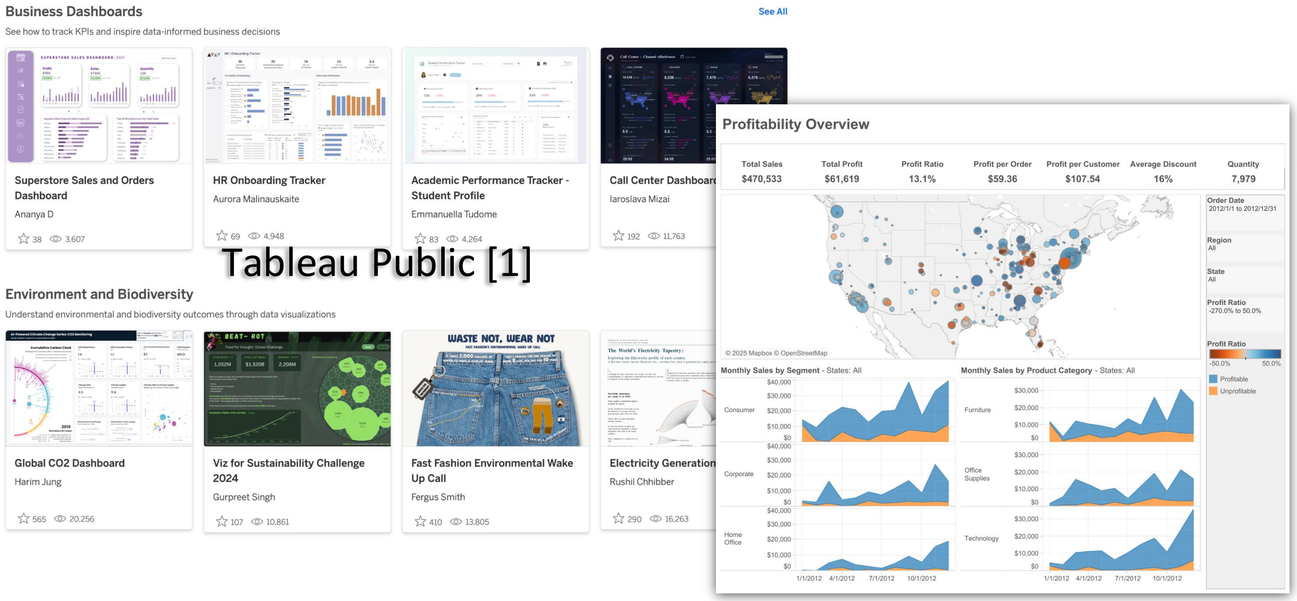


Tableau Public [1]

🏆 Awarded Entry in 2021 IEEE VAST Challenge
Mini Challenge 3

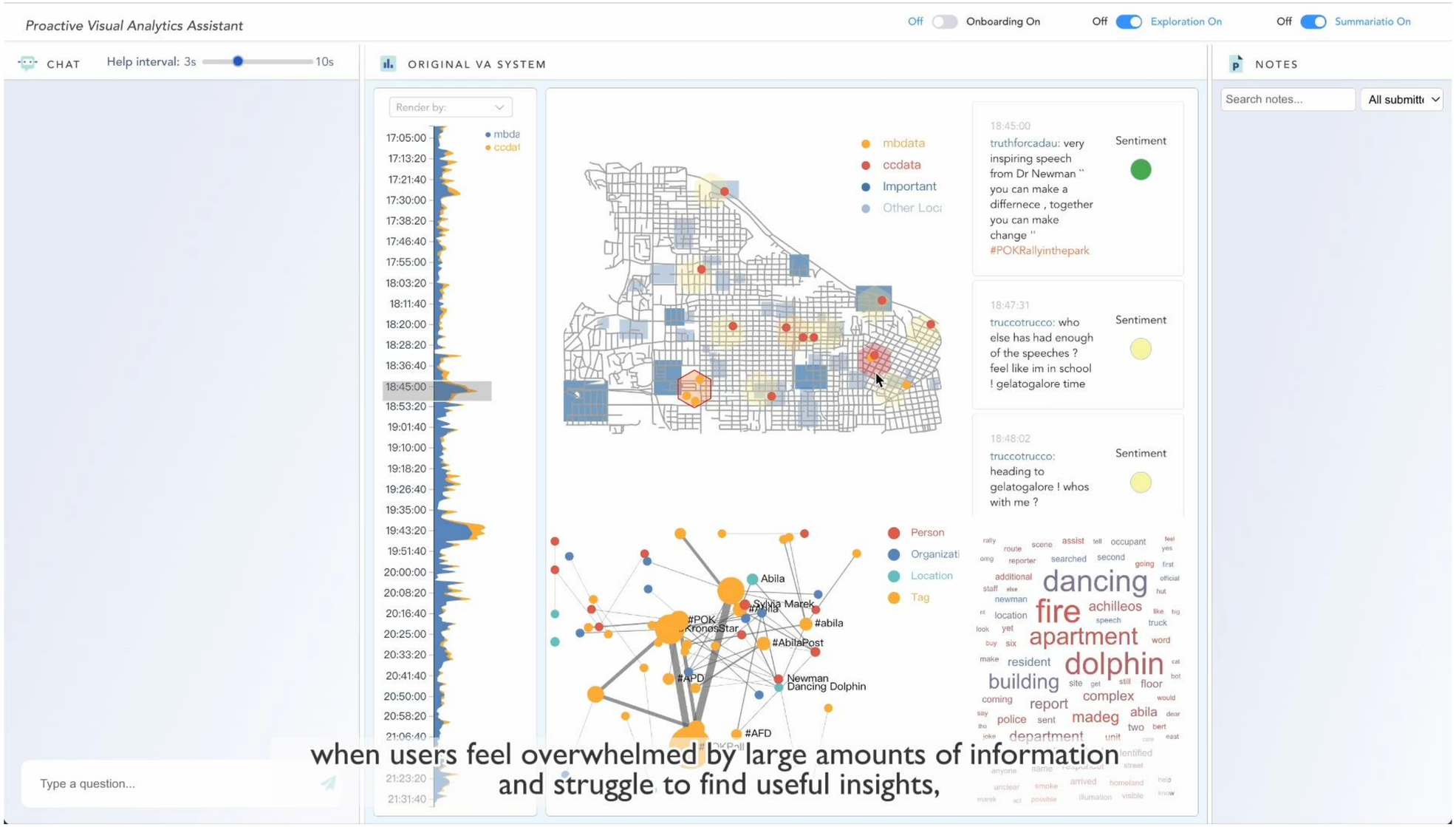
Data: Social media data
Task: Identify events and evaluate risks in Abila City

📈 Classic Sales and Profit Analysis
Tableau Dashboard

Data: Tabular data
Task: Analyze the sales profit of the product
Adaptation: Tableau Embedding API [2]

[1] Tableau Public https://public.tableau.com/app/discover
[2] Tableau Embedding API. https://help.tableau.com/current/api/embedding_api/en-us/index.html

Exploration Guidance - IEEE VAST Challenge 2021



Full video at: https://zyh1222.github.io/ProactiveVA/, *Vast Challenge and Tableau*

Evaluation

1. Algorithm Evaluation:

Batch Simulation Approach: Task generation - Automated sandbox execution - Scoring

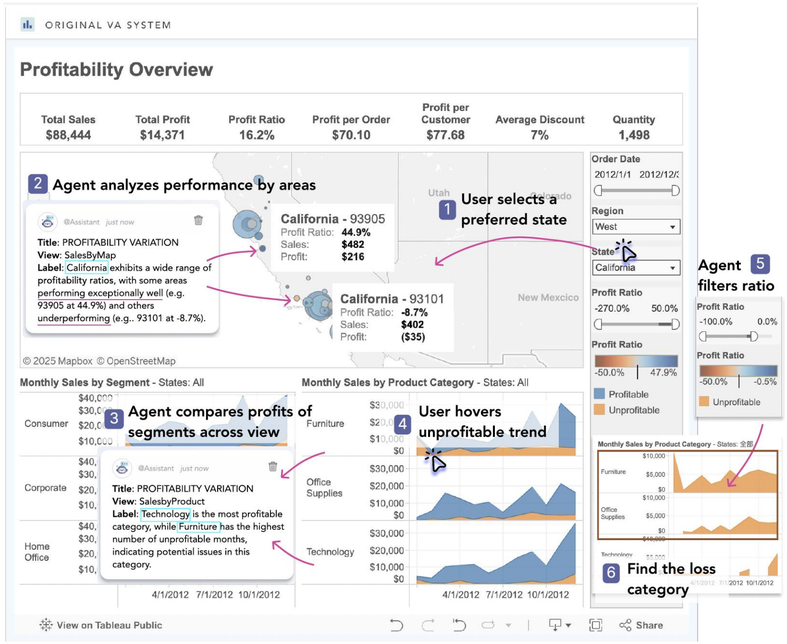
Table 1: Evaluation of model performance across different data analysis tasks and metrics.

Category	Count	Time (s) Mean (±Std)	Step Count Mean (±Std)	LLM Evaluation			Human Evaluation		
				Task Com.	Data Acc.	Path Eff.	Task Com.	Data Acc.	Path Eff.
Total/Avg.	100	35.94 (±13.47)	4.36 (±1.61)	4.15	4.96	4.45	4.38	4.85	4.67
Comparison	17	40.81 (±10.44)	5.12 (±1.41)	4.06	5.00	4.24	4.41	4.94	4.65
Trend	20	31.90 (±9.49)	3.80 (±1.03)	4.20	5.00	4.70	4.45	4.85	4.80
Performance	31	33.72 (±8.90)	3.94 (±0.88)	4.16	4.87	4.52	4.42	4.81	4.71
Correlation	11	33.81 (±11.77)	4.36 (±1.77)	3.64	5.00	4.18	3.73	4.64	4.27
Dimension	21	40.22 (±20.93)	4.90 (±2.37)	4.43	5.00	4.43	4.57	4.95	4.71

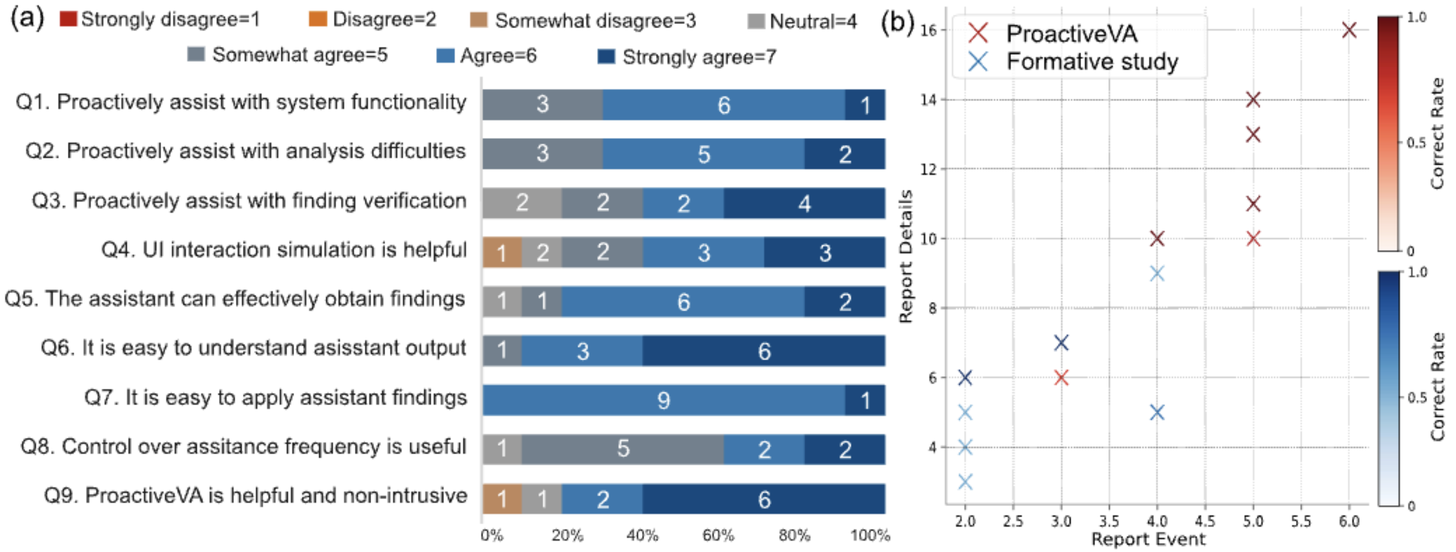
Key findings

- The UI agent has self-correction capabilities through trial and error
- The exploration paths and parameter settings align with expert habits.
- Resolving misunderstandings and errors
- It shows improvements in depth, breadth, and accuracy and lowering the threshold for beginners.

2. Expert Study:



3. User Study:



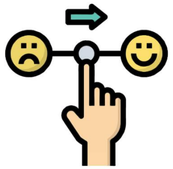
Future work





Optimizing intent understanding:

- Handle complex, long-horizon task planning (Hierarchical modeling)
- Improve visual grounding to better handle noise and ambiguity (VLMs)



Adaptive Proactivity

- Tune the frequency, granularity, and intensity of proactivity based on user's analytical literacy



Interactive and Verifiable Methods

- Surface system's uncertainties and enable provenance for transparency and trust

FDUVIS Lab

Thank you!

yuhengzhao@fudan.edu.cn
simingchen@fudan.edu.cn

More details in our paper👀

Project Website: https://zyh1222.github.io/ProactiveVA/

VIS2025

